23EGMO1

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Problem

There are $n \geq 3$ positive real numbers $a_1, a_2, \ldots, a_n$. For each $1 \leq i \leq n$ we let $b_i = \dfrac{a_{i-1} + a_{i+1}}{a_i}$ (here we define a_0 to be a_n and a_{n+1} to be a_1). Assume that for all i and j in the range 1 to n, we have $a_i \leq a_j$ if and only if $b_i \leq b_j$. Prove that $a_1 = a_2 = \cdots = a_n$.

Solution. Let $a_m = \min\limits_i a_i$ and $a_M = \max\limits_i a_i$. This means that $b_m = \min\limits_i b_i$ and $b_M = \max\limits_i b_i$.

Now, $b_m = \dfrac{a_{m-1} + a_{m+1}}{a_m} \geq 2$ and $b_M = \dfrac{a_{M-1} + a_{M+1}}{a_M} \leq 2$.

Since, for all $1 \leq k \leq n$, $2 \leq b_m \leq b_k \leq b_M \leq 2$, this implies that $b_k = 2$ for all $1 \leq k \leq n$.

As a result, $a_i - a_{i-1} = a_{i+1} - a_i$, so $\{a_k\}$ in an arithmetic progression, say with common difference d.

So, $a_n = a_1 + (n-1)d$ and $a_n - a_{n-1} = d = a_{n+1} - a_n = a_1 - a_n = -(n-1)d$. This of course means that $d = 0$ since $n \geq 3$.

As a result, $a_1 = a_2 + \cdots = a_n$. $\square$